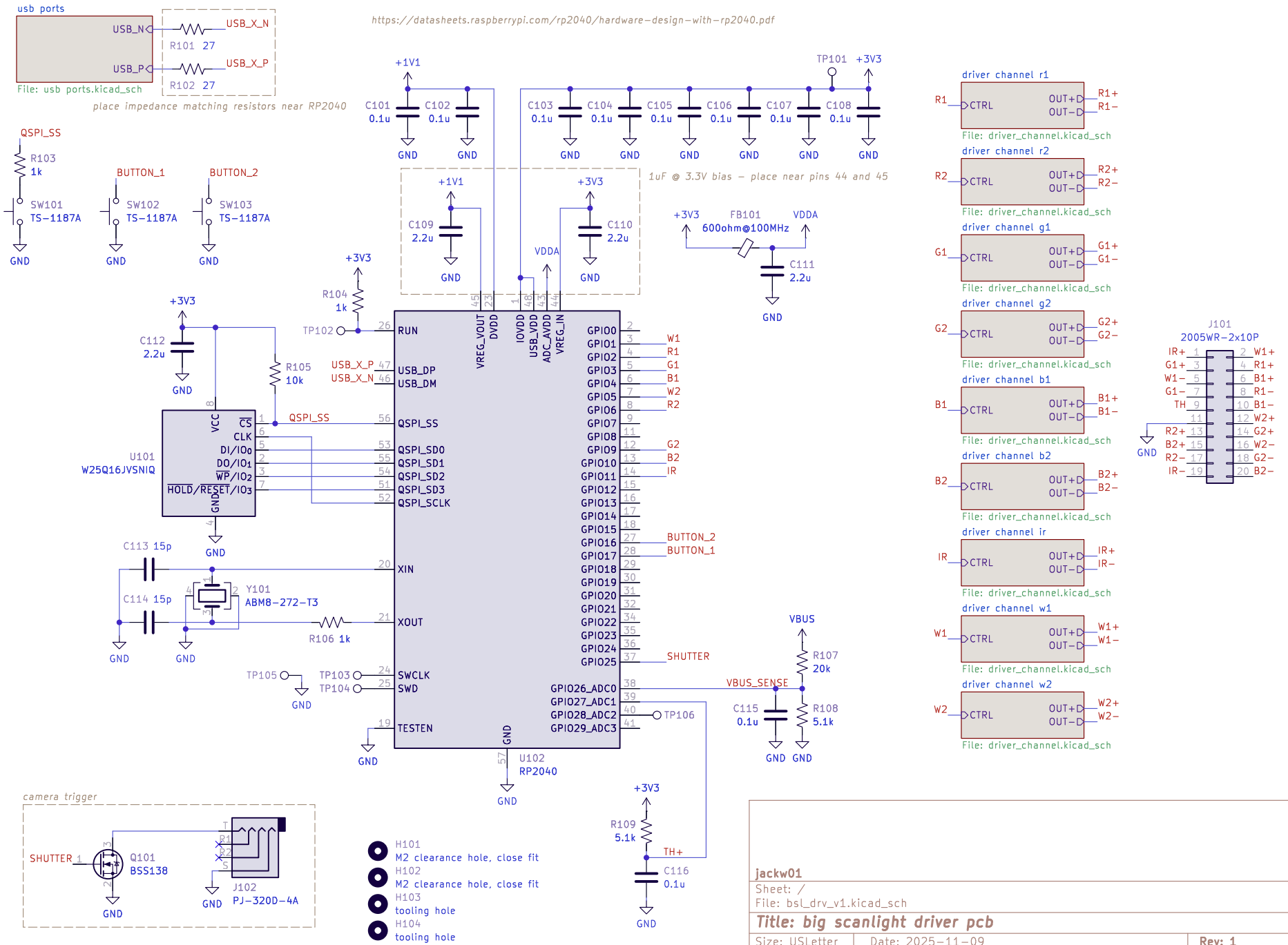
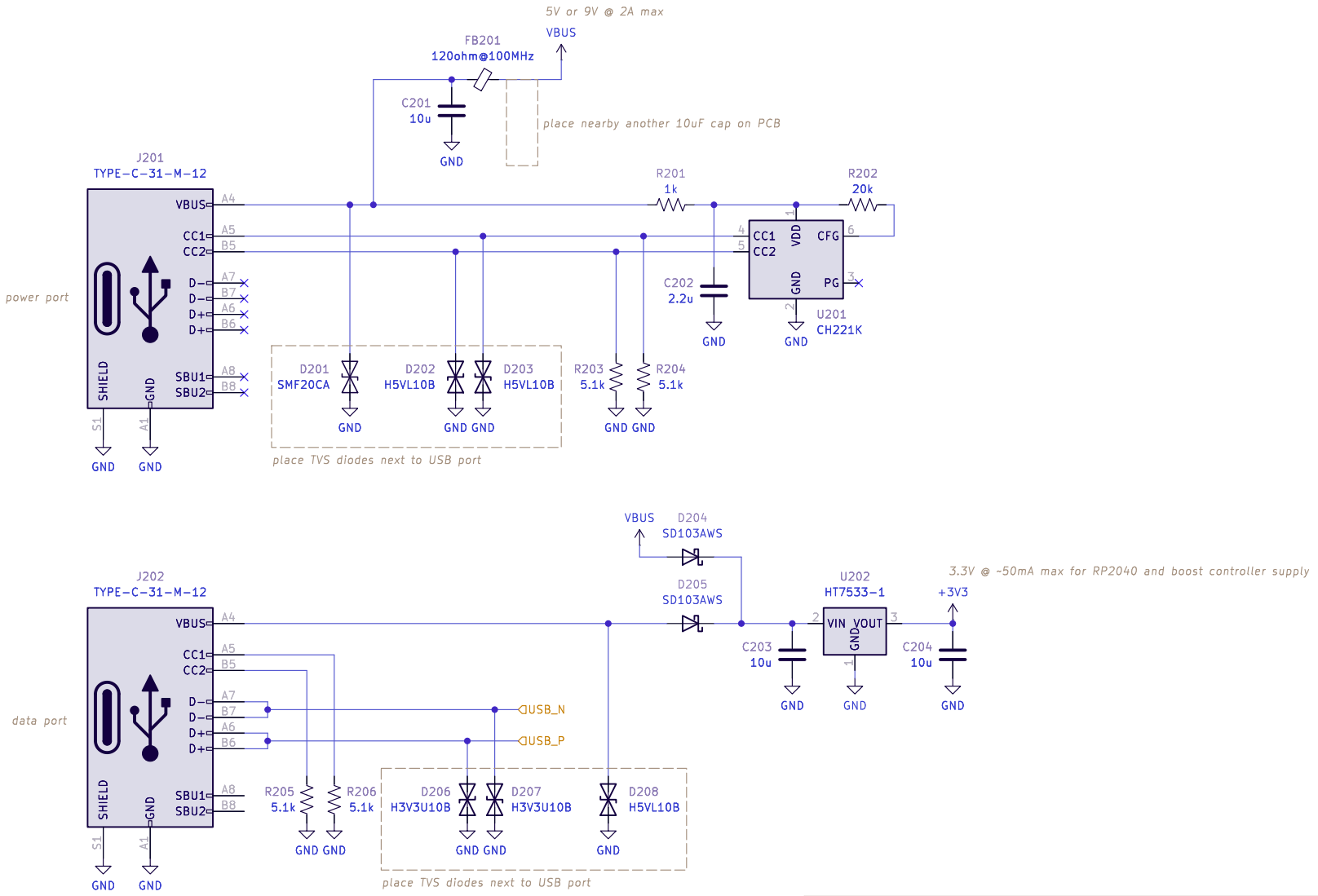
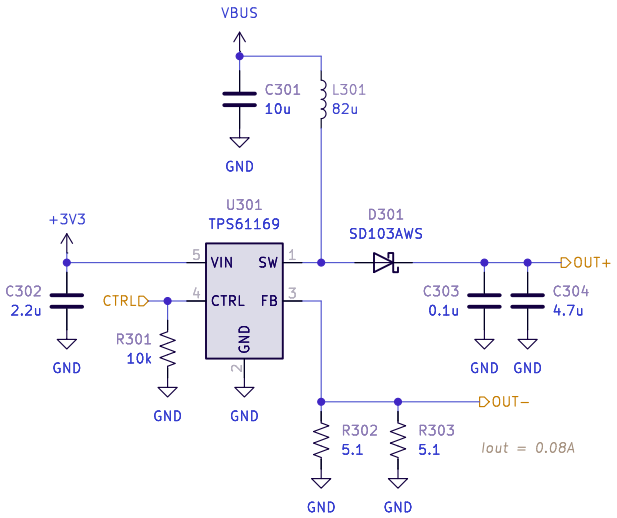
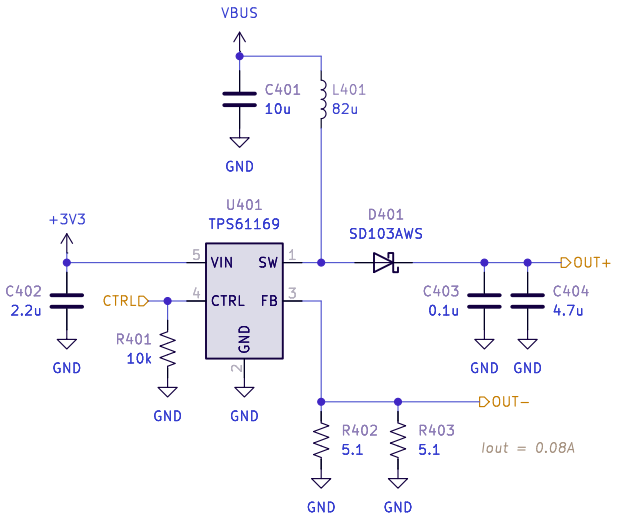

usb ports
USB_N
USB_P
USB_X_N
USB_X_P
R101 27
R102 27
File: usb ports.kicad_sch
place impedance matching resistors near RP2040
https://datasheets.raspberrypi.com/rp2040/hardware-design-with-rp2040.pdf
QSPI_SS
R103 1k
SW101 TS-1187A
SW102 TS-1187A
SW103 TS-1187A
BUTTON_1
BUTTON_2
+1V1
C101 0.1u
C102 0.1u
C103 0.1u
C104 0.1u
C105 0.1u
C106 0.1u
C107 0.1u
C108 0.1u
TP101
+3V3
GND
1uF @ 3.3V bias - place near pins 44 and 45
C109 2.2u
C110 2.2u
VDDA
FB101 600ohm@100MHz
C111 2.2u
R104 1k
TP102
C112 2.2u
R105 10k
U101
W25Q16JVSNIQ
VCC
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
C113 15p
C114 15p
Y101 ABM8-272-T3
R106 1k
TP105
TP103
TP104
SWCLK
SWD
TESTEN
XIN
XOUT
U102
RP2040
RUN
USB_DP
USB_DM
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
VREG_VOUT
DVDD
IOVDD
USB_VDD
ADC_AVDD
VREG_IN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
W1
R1
G1
B1
W2
R2
G2
B2
IR
BUTTON_2
BUTTON_1
SHUTTER
TP106
VBUS
R107 20k
R108 5.1k
C115 0.1u
VBUS_SENSE
R109 5.1k
TH+
C116 0.1u
camera trigger
SHUTTER
Q101 BSS138
J102 PJ-320D-4A
H101 M2 clearance hole, close fit
H102 M2 clearance hole, close fit
H103 tooling hole
H104 tooling hole
driver channel r1
driver channel r2
driver channel g1
driver channel g2
driver channel b1
driver channel b2
driver channel ir
driver channel w1
driver channel w2
CTRL
OUT+
OUT-
File: driver_channel.kicad_sch
J101
2005WR-2x10P
jackw01
Sheet: /
File: bsl_drv_v1.kicad_sch
Title: big scanlight driver pcb
Size: USLetter
Date: 2025-11-09
Rev: 1
KiCad E.D.A. 9.0.1-rc2
Id: 1/11



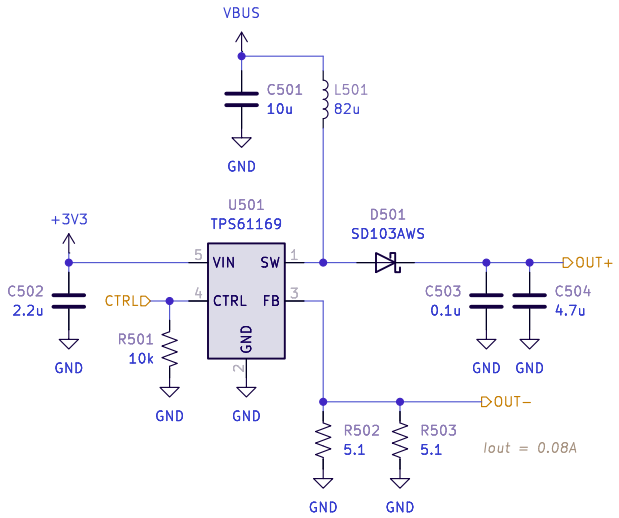
jackw01	
Sheet: /usb ports/	
File: usb ports.kicad_sch	
Title: big scanlight driver pcb	
Size: USLetter | Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2	Id: 2/11



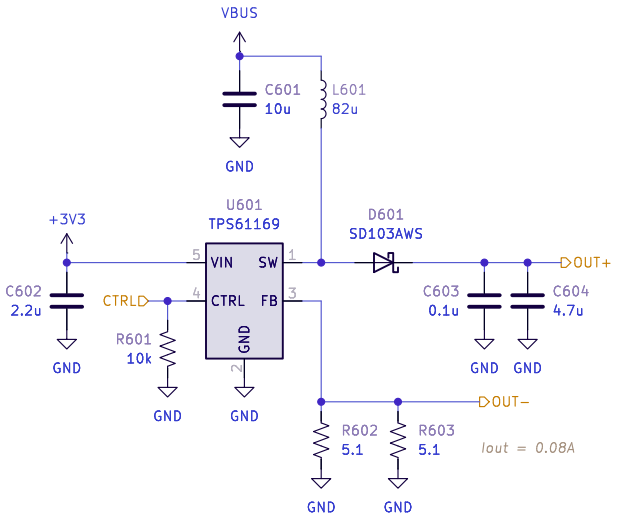
jackw01	
Sheet: /driver channel r1/	
File: driver_channel.kicad_sch	
Title: big scanlight driver pcb	
Size: USLetter | Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2	Id: 3/11



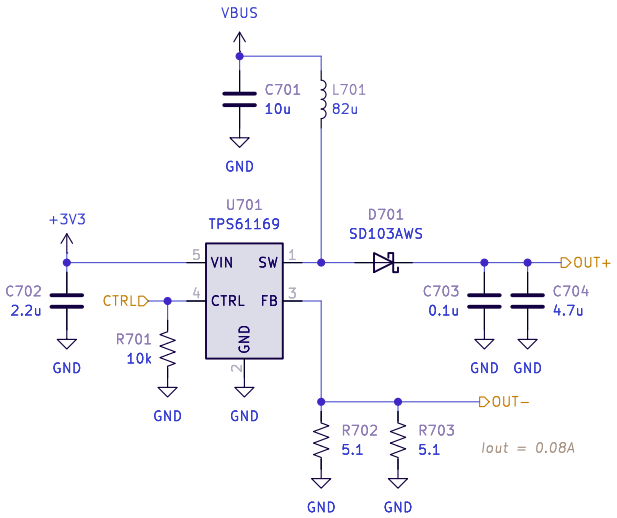
jackw01		
Sheet: /driver channel r2/		
File: driver_channel.kicad_sch		
Title: big scanlight driver pcb		
Size: USLetter	Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2		Id: 4/11



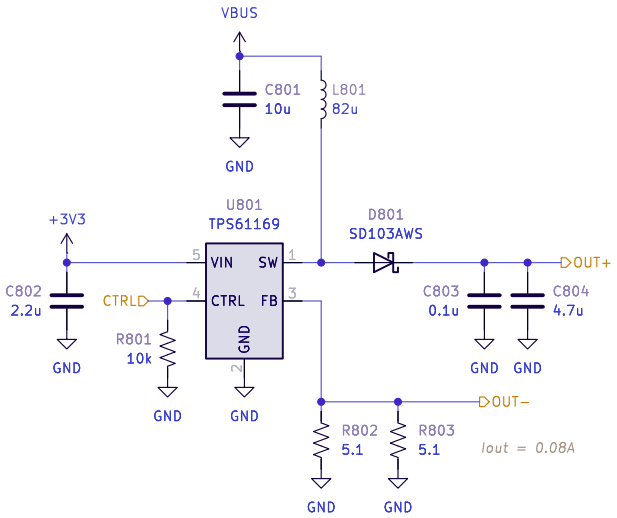
jackw01	
Sheet: /driver channel g1/	
File: driver_channel.kicad_sch	
Title: big scanlight driver pcb	
Size: USLetter | Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2	Id: 5/11



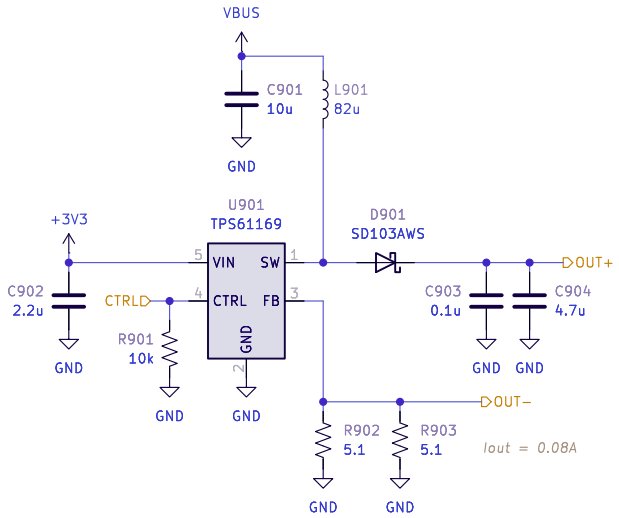
jackw01	
Sheet: /driver channel g2/	
File: driver_channel.kicad_sch	
Title: big scanlight driver pcb	
Size: USLetter | Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2	Id: 6/11



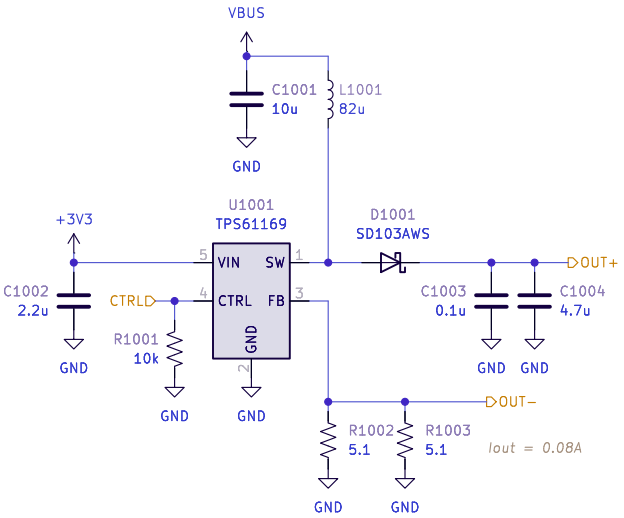
jackw01	
Sheet: /driver channel b1/	
File: driver_channel.kicad_sch	
Title: big scanlight driver pcb	
Size: USLetter | Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2	Id: 7/11



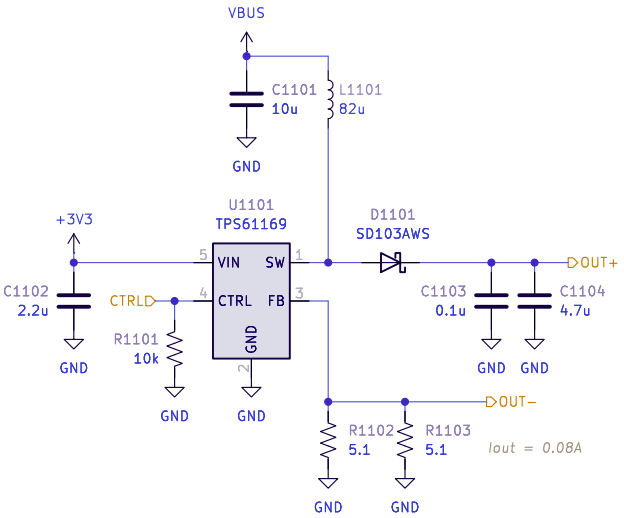
jackw01		
Sheet: /driver channel b2/		
File: driver_channel.kicad_sch		
Title: big scanlight driver pcb		
Size: USLetter	Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2		Id: 8/11



jackw01		
Sheet: /driver channel ir/		
File: driver_channel.kicad_sch		
Title: big scanlight driver pcb		
Size: USLetter	Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2		Id: 9/11



jackw01		
Sheet: /driver channel w1/		
File: driver_channel.kicad_sch		
Title: big scanlight driver pcb		
Size: USLetter	Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2		Id: 10/11



jackw01		
Sheet: /driver channel w2/		
File: driver_channel.kicad_sch		
Title: big scanlight driver pcb		
Size: USLetter	Date: 2025-11-09	**Rev: 1**
KiCad E.D.A. 9.0.1-rc2		Id: 11/11